

VLSI INTERNSHIP – TASK 4

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Domain: Very Large Scale Integration (VLSI) Intern

Goal

Introduce Design for Testability (scan insertion) and understand layout vs schematic checks (LVS/DRC) at a conceptual or tool-driven level.

Requirements

- Apply basic scan chain insertion using DFT tool or describe manual approach.
- Explain LVS/DRC steps and, if tools available, run checks on a sample layout.
- Discuss test coverage and stuck-at faults.

8-bit ALU with scan chain code

```
module alu8bit_scan (  
    input clk,  
    input rst,  
    input scan_en,  
    input scan_in,  
    output scan_out,  
    input [7:0] A,  
    input [7:0] B,  
    input [2:0] sel,  
    output reg [7:0] Y);  
    reg [7:0] alu_result;  
    always @(*) begin  
        case(sel)  
            3'b000: alu_result = A + B;  
            3'b001: alu_result = A - B;  
            3'b010: alu_result = A & B;  
            3'b011: alu_result = A | B;  
            3'b100: alu_result = A ^ B;  
            3'b101: alu_result = ~A;  
            3'b110: alu_result = A << 1;  
            3'b111: alu_result = A >> 1;  
            default: alu_result = 8'b0;  
        endcase  
    end  
    always @(posedge clk or posedge rst) begin  
        if (rst)  
            Y <= 8'b0;  
        else if (scan_en)  
            Y <= {Y[6:0], scan_in};  
        else  
            Y <= alu_result;  
        end  
    assign scan_out = Y[7];  
endmodule
```

Testbench

```
`timescale 1ns/1ps

module alu8bit_scan_tb;
reg clk;
reg rst;
reg scan_en;
reg scan_in;
reg [7:0] A, B;
reg [2:0] sel;
wire scan_out;
wire [7:0] Y;
alu8bit_scan uut (.clk(clk),.rst(rst),.scan_en(scan_en),.scan_in(scan_in),.scan_out(scan_out),.A(A),.B(B),.sel(sel),.Y(Y));
always #5 clk = ~clk;
initial begin
    $dumpfile("alu8bit_scan.vcd");
    $dumpvars(0, alu8bit_scan_tb);
    clk = 0;
    rst = 1;
    scan_en = 0;
    scan_in = 0;
    A = 0;
    B = 0;
    sel = 0;
    #10;rst = 0;
    $display("Functional Mode");
    A = 8'd10; B = 8'd5; sel = 3'b000;
    #10;

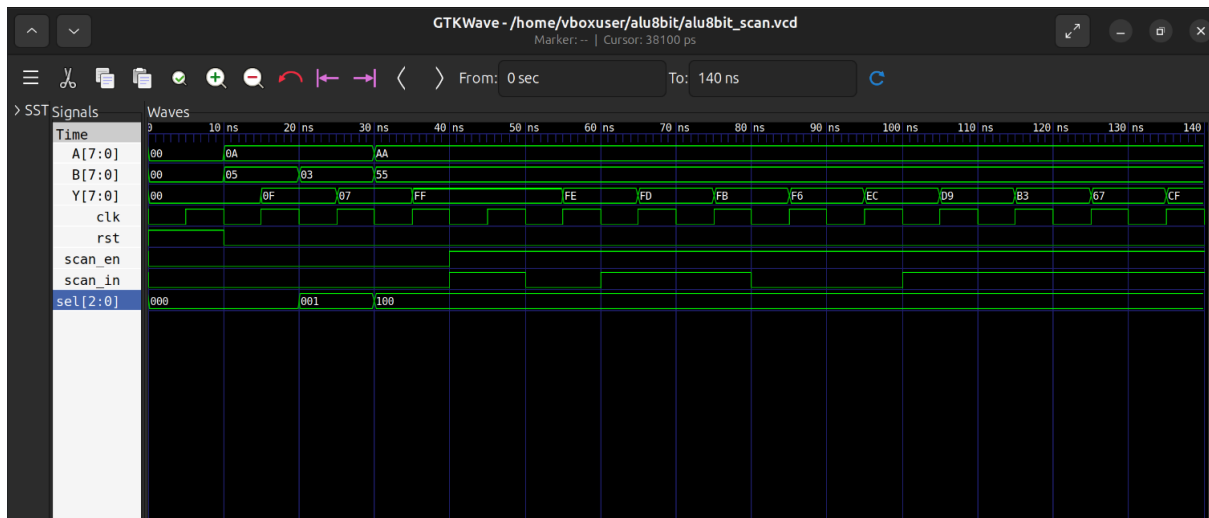
    $display("ADD: Y = %d", Y);
    A = 8'd10; B = 8'd3; sel = 3'b001;
    #10;
    $display("SUB: Y = %d", Y);
    A = 8'hAA; B = 8'h55; sel = 3'b100;
    #10;
    $display("XOR: Y = %h", Y);
    $display(" Scan Mode ");
    scan_en = 1;

    scan_in = 1; #10;
    scan_in = 0; #10;
    scan_in = 1; #10;
    scan_in = 1; #10;
    scan_in = 0; #10;
    scan_in = 0; #10;
    scan_in = 1; #10;
    scan_in = 1; #10;

    $display("Scan Register Y = %b", Y);
    $display("Scan Out = %b", scan_out);

    #20;
    $finish;
end
```

GTKWave



Commands

```
vboxuser@ubuntu:~/alu8bit$ iverilog -o alu8bit_scan_out alu8bit_scan.v alu8bit_scan_tb.v
vboxuser@ubuntu:~/alu8bit$ vvp alu8bit_scan_out
VCD info: dumpfile alu8bit_scan.vcd opened for output.
Functional Mode
ADD: Y = 15
SUB: Y = 7
XOR: Y = ff
Scan Mode
Scan Register Y = 10110011
Scan Out = 1
alu8bit_scan_tb.v:77: $finish called at 140000 (1ps)
```

The design contains 8 flip-flops corresponding to the 8-bit output register Y. A single scan chain can be formed by replacing standard DFFs with scan DFFs and connecting them serially. During test mode (`scan_en=1`), test patterns are shifted through the scan chain to improve controllability and observability.

DRC (Design Rule Check):

```
alu8bit
-----
[INFO]: COUNT: 0
[INFO]: Should be divided by 3 or 4
```

Design Rule Check (DRC) was performed on the final layout generated by OpenLane using the Sky130A technology. The DRC report indicates 0 violations, confirming that the layout satisfies all required design rules, including spacing, width, enclosure, and overlap constraints.

LVS (Layout Versus Schematic)

LVS reports no net, device, pin, or property mismatches.

Total errors = 0

No net mismatches

No device mismatches

No pin mismatches

No property mismatches

Total Errors = 0

Physical Layout

